

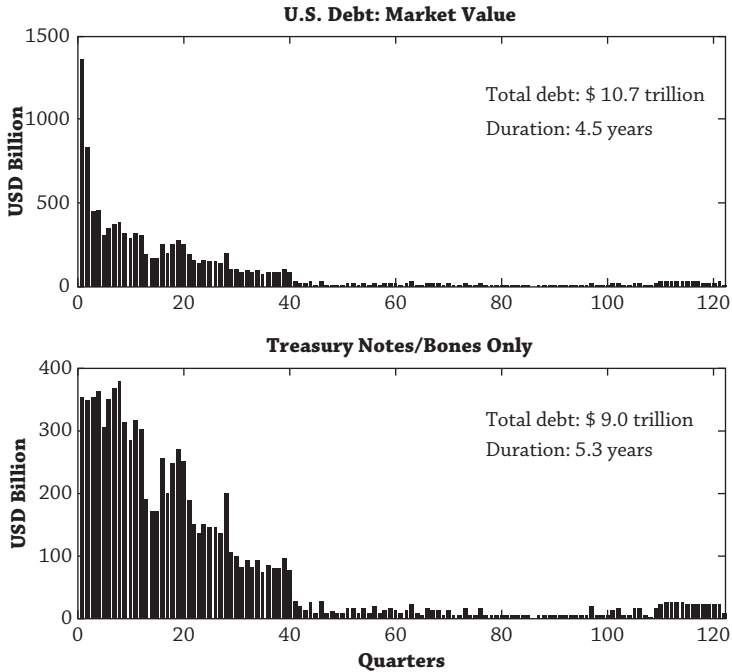


Figure 14.12

issues most of its debt with short maturities.⁵¹ Do you really want a maturity structure like Figure 14.12 if you are bound to market weights?

The level factor, or the average interest rate across maturities, is by far the most important factor for fixed income (see chapter 9). The level factor accounts for 90% or more of the variation of safe asset returns and yields.⁵² We could create a level factor portfolio quite easily by taking select bonds across the maturity structure. Exposure to the level factor is labeled *duration*, and once the level factor is constructed, an investor selects an appropriate duration target.⁵³

An objection might be: but if you select a few bonds (somewhat arbitrarily), aren't you missing out on a lot of issues? Yes. But a dirty secret is that most investors never hold all the bonds in fixed income indexes. In many cases, it is impossible to hold all the securities because some issues just don't trade or are

⁵¹ There are few theories on how a government should select a maturity structure for its debt. The leading paper is Angeletos (2002), who shows how to derive an optimal maturity structure that ensures against fiscal shocks and allows the government to smooth taxes.

⁵² See Litterman and Scheinkman (1991).

⁵³ Term structure movements across countries are highly correlated, as Kose, Otrok, and Whiteman (2003) and Jotikasthira, Le, and Lundblad (2010) show. Setting one duration target across all countries is a reasonable thing to do.